Kinect Lite: Remaining Work

Status as of 2026-10-24

Done

- Three-step spatial pipeline: 0.4 m gate, median filter, inpainting.
- Tuning and evaluation on ICL-NUIM: error 34.7 mm to 13.0 mm, every frame improved.
- TUM diagnostics, figures, 3D viewers, RESULTS/, docs and progress report.

Still to do: Phase 2 (ICP), not started

1. **Implement ICP** (point-to-point or point-to-plane) on back-projected clouds; denoise/pointcloud.py already does back-projection.
2. **Load ground-truth poses.** ICL: livingRoom0.gt.freiburg (ids start at 1, depth at 0; check offset). TUM: groundtruth.txt, 8710 poses, associate by nearest timestamp.
3. **Run ICP on raw vs cleaned depth** (e.g. iamedian k9 + 0.4 m gate, optionally hole-filled).
4. **Compare trajectories with ground truth** using ATE and RPE, to see if denoising improves tracking on the real sensor.
5. **Write up** results in the README, RESULTS/ and the progress report.

Things to watch

- ICL noisy vs ground truth is misaligned by about 1 px (not corrected).
- TUM has about 16.5% missing depth, which ICP must handle.
- ICL fy is listed as -480 (y-up): check the sign when back-projecting.

Housekeeping

- Uncommitted change in .gitignore.
- CHECKPOINT_SPATIAL.md mentions REPORTS/ and denoise_code.zip, but the repo has kinect_lite_pipeline.zip; checkpoint may be stale.